



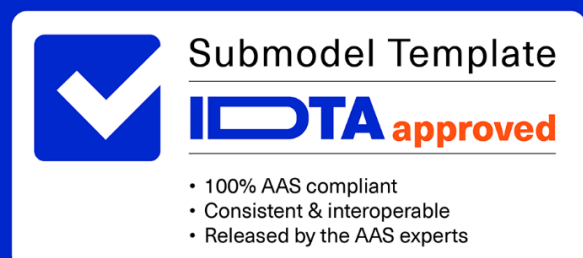
IDTA 02035-5 : Digital Battery Passport - Part 5 Product Condition

Version 1.0.1

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SPECIFICATION

Submodel Template of the
Asset Administration Shell



IDTA 02035-5 V1.0.1

Imprint

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For details see Changelog in Annex.

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18.02.2026	1.0	Release of the official Submodel Template published by IDTA – a joint result of the Model Expert Group comprising IDTA, Catena-X, and the Battery Pass Consortium.

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Chapter 1. General

1.1. About this document

This document is a part of an overall specification series [4]. Each part specifies the contents of a Submodel Template (SMT). The specifications of the Asset Administration Shell (AAS) are the basis for the Submodel Template specifications, see [3].

The target audience of the specification are developers and editors of technical documentation and manufacturer information, which are describing assets by means of the Asset Administration Shell (AAS) and therefore need to create a Submodel instance with a hierarchy of SubmodelElements. This document especially details on the question, which SubmodelElements with which semantic identification shall be used for this purpose.

This specification was created following the "semantic-driven workflow" as defined in [5] based on Aspect Models [6]. There is no central dictionary or repository for Aspect Models. In this specification the following sources are used for defining semantics:

- Aspect Models published at IDTA [7]: <https://github.com/admin-shell-io/smt-semantic-models>, models with namespace "io.admin-shell"
- Aspect Models published at BatteryPass Consortium (closed project) [8]: <https://github.com/batterypass/BatteryPassDataModel>, models with namespace "io.BatteryPass"
- Aspect Models published at Tractus-X and used in standards published by Catena-X [9]: <https://github.com/eclipse-tractusx/sldt-semantic-models>, models with namespace "io.catenax"

1.2. Scope of the Submodel

This Submodel template aims to define the dynamic data points of a battery item required to be included in a Battery Passport conformant to DIN DKE SPEC 99100 and the corresponding EU regulations.

The battery passport consists of the following 7 parts:

Digital Battery Passport - Part 1: Digital Nameplate (IDTA-02035-1)
Digital Battery Passport - Part 2: Handover Documentation (IDTA-02035-2)
Digital Battery Passport - Part 3: Product Carbon Footprint (IDTA-02035-3)
Digital Battery Passport - Part 4: Technical Data (IDTA-02035-4)
Digital Battery Passport - Part 5: Product Condition (IDTA-02035-5)
Digital Battery Passport - Part 6: Material Composition (IDTA-02035-6)
Digital Battery Passport - Part 7: Circularity (IDTA-02035-7)

This specification is Part 5: Product Condition (IDTA_02035-5).

1.3. Relevant standards for the Submodel Template

This submodel template fulfills the requirements for dynamic data points of a battery item as defined in DIN DKE SPEC 99100 [1]. DIN DKE SPEC 99100 "is based on the European Union and key Member States current regulatory requirements for battery passport information. Mandatory information for the battery passport as stated in the EU Battery Regulation (EU)2023/1542, Article77 and AnnexXIII, as well as the Ecodesign for Sustainable Products Regulation (ESPR), is supplemented by recommendations to increase sustainability and circularity. [1]"

This document is valid for all battery categories. Please be aware that for battery categories that have stronger requirements like industrial batteries with battery management systems etc. some of the data points are specified as optional although mandatory per regulation.

1.4. Explanations on used UML diagrams

For clarity and an improved legibility readers suggested to go through this section at first before reading the following chapters.

UML diagrams feature box-like elements, called "classes". These classes, typically Submodels, SubmodelElementCollections or SubmodelElementLists, typically feature a set of Properties or further SubmodelElements. These elements can have specific cardinalities.

The single classes are hierarchally organized by aggregation relations, these can be seen as "contains" relation.

For a further overview on UML diagrams please refer to [2] and [3].

Further details about used table formats please refer to Annex A Explanations.

Chapter 2. Information set for Submodel “ProductCondition”

2.1. General

The "Product Condition" Submodel Template is part of the specification series for the Battery Passport.

Property specification

See [clause 3 "Information structures and attributes"](#).

2.2. Overview UML model

The SubmodelElements described in [clause 3 "Information structures and attributes"](#) are structured in the following way (see [Figure 1](#)):

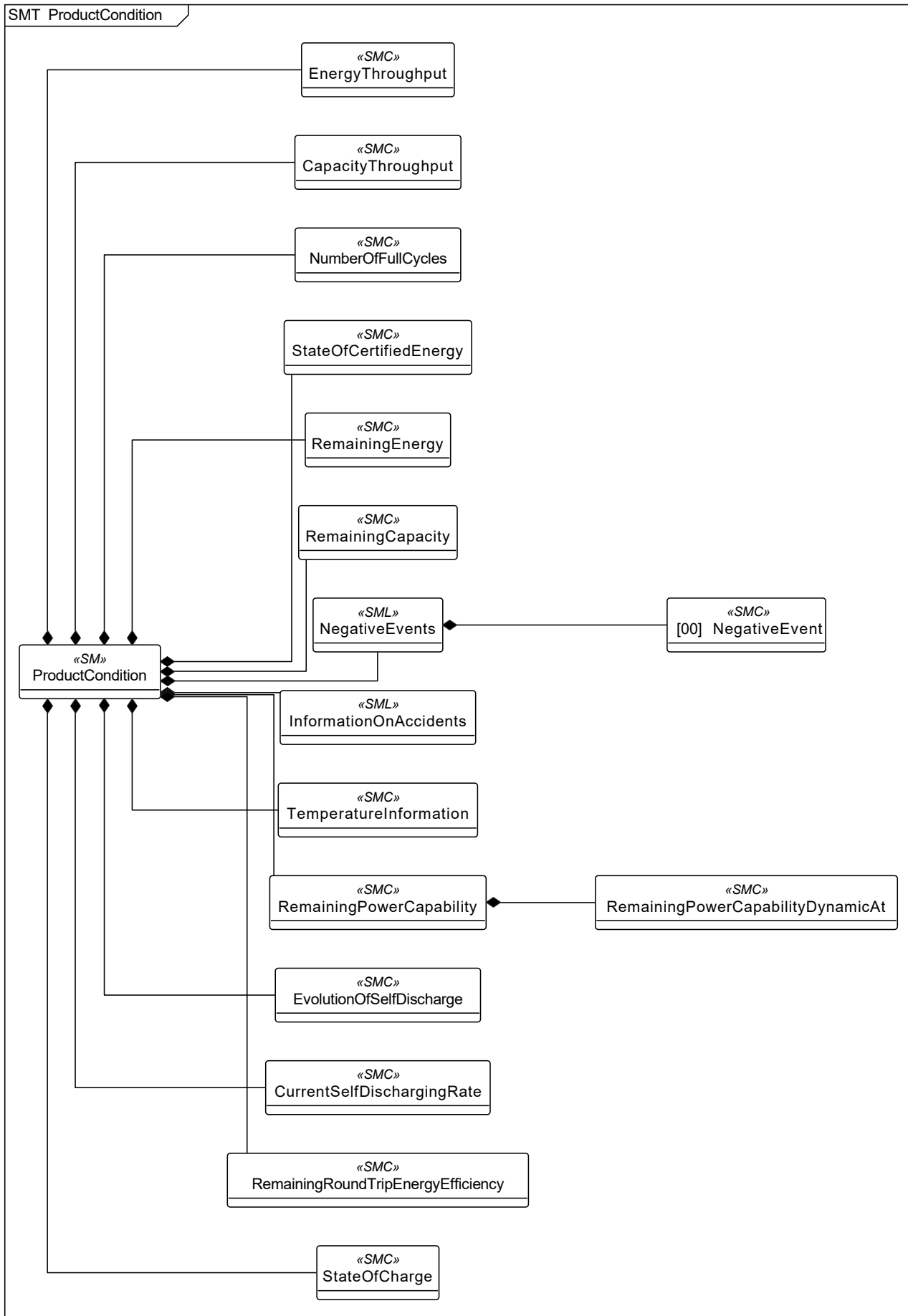


Figure 1. UML overview

The following design principles were followed:

- for all dynamic data points there is an attribute "lastUpdate" showing the last update time

Chapter 3. Information structures and attributes

3.1. Properties of the Submodel “ProductCondition”

Figure 2 shows the UML-diagram defining the relevant properties which need to be set.

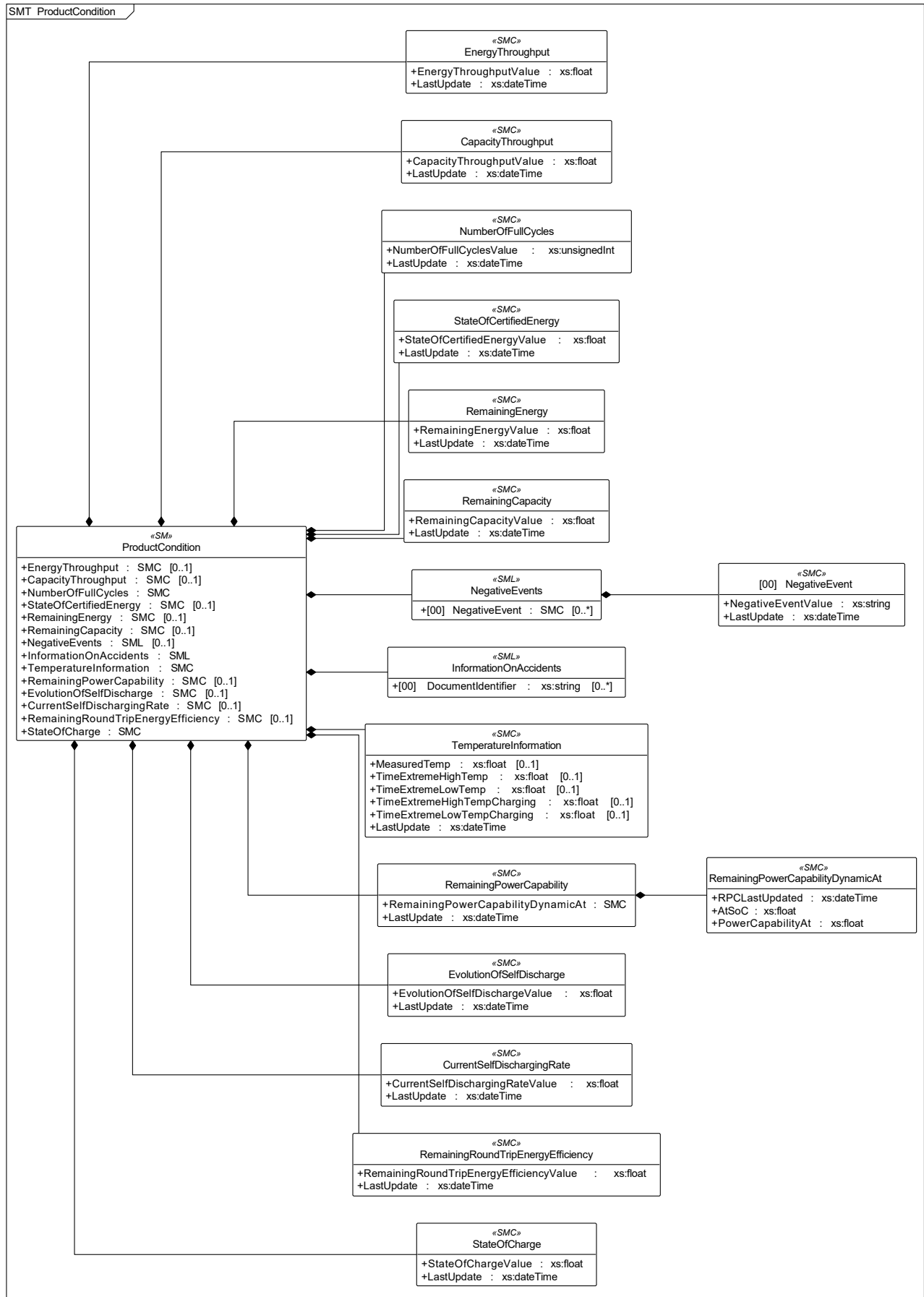


Figure 2. UML-Diagram for Submodel "ProductCondition"

idShort:	ProductCondition
Class:	Submodel

semanticId:	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#ProductCondition		
Parent:	-		
Explanation:	Covers all battery lifetime relevant properties.		
Element details:	-		
[SME type]	semanticId	[valueType]	card.
idShort	Description@en	example	
[SMC] EnergyThroughput	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#energyThroughput supplementalSemanticId: urn:irdi:0173-1#02-ABL850#001 The data attribute should be reported as measured for further potential processing. In addition, the normalisation by capacity may add a further useful value that ensures comparability among battery sizes. DIN DKE Spec 99100 chapter reference: 6.7.6.7	[] 2 elements	0..1
[SMC] CapacityThroughput	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#capacityThroughput supplementalSemanticId: urn:irdi:0173-1#02-ABL849#001 The data attribute should be reported as measured for further potential processing. In addition, the normalisation by capacity may add a further useful value that ensures comparability among battery sizes. DIN DKE Spec 99100 chapter reference: 6.7.6.8	[] 2 elements	0..1
[SMC] NumberOfFullCycles	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#numberOfFullCycles supplementalSemanticId: urn:irdi:0173-1#02-ABL868#001 Number of (full) charging and discharging cycles. DIN DKE Spec 99100 chapter reference: 6.7.6.4	[] 2 elements	1

[SMC] StateOfCertifiedEnergy	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#stateOfCertifiedEnergy supplementalSemanticId: urn:irdi:0173-1#02-ABL847#001 Definition based on UNECE GTR 22: The measured or on-board UBE performance at a specific point in its lifetime, expressed as a percentage of the certified usable battery energy. DIN DKE Spec 99100 chapter reference: 6.7.2.7	[] 2 elements	0..1
[SMC] RemainingEnergy	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#remainingEnergy supplementalSemanticId: urn:irdi:0173-1#02-ABL870#001 Definition from UNECE GTR 22, applicable only to EVs. The energy supplied by the battery from the beginning of the test procedure used for certification until the applicable break-off criterion of the test procedure used for certification is reached. DIN DKE Spec 99100 chapter reference: 6.7.2.6	[] 2 elements	0..1
[SMC] RemainingCapacity	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#remainingCapacity supplementalSemanticId: urn:irdi:0173-1#02-ABL840#001 The in-use data attribute on capacity, corresponding with the definition of rated capacity. DIN DKE Spec 99100 chapter reference: 6.7.2.3	[] 2 elements	0..1
[SML] NegativeEvents	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#negativeEvents The battery passport must contain information and data resulting from its use such as accidents. DIN DKE Spec 99100 chapter reference: 6.7.8.4	[] 1 elements	0..1

[SML] InformationOn Accidents	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#informationOnAccidents supplementalSemanticId: urn:samm:io.admin-shell.idta.shared:3.1.0#DocumentIdentifierSet, urn:irdi:0173-1#02-ABL848#001 The battery passport must contain information and data resulting from its use such as accidents. DIN DKE Spec 99100 chapter reference: 6.7.8.4	[] 1 elements	1
[SMC] TemperatureIn formation	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#temperatureInformation The battery passport must include periodically recorded information on the operating environmental conditions, including temperature. DIN DKE Spec 99100 chapter reference: 6.7.7.5 - 8	[] 6 elements	1
[SMC] RemainingPow erCapability	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#remainingPowerCapability supplementalSemanticId: urn:irdi:0173-1#02-ABL820#001 Original power capability (in Watts) and limits, with temperature range when relevant. - The amount of energy that a battery is capable to provide over a given period of time under reference conditions. - Power capability at 80% and 20% state of charge. DIN DKE Spec 99100 chapter reference: 6.7.3.3	[] 2 elements	0..1
[SMC] EvolutionOfSel fDischarge	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#evolutionOfSelfDischarge Evolution of self-discharge rates is the change of self-discharge over time and usage, as percentage calculated from the initial and current self-discharge rate. DIN DKE Spec 99100 chapter reference: 6.7.4.8	[] 2 elements	0..1
[SMC] CurrentSelfDis chargingRate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#currentSelfDischargingRate The current self-discharge rate is the change of the self-discharge rate in an idle state of the battery in reference conditions (temperature etc.) at aging parameter x, e.g. after a certain amount of storage time or, number of cycles. DIN DKE Spec 99100 chapter reference: 6.7.4.7	[] 2 elements	0..1

[SMC] RemainingRoundTripEnergyEfficiency	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#remainingRoundTripEnergyEfficiency The battery passport shall include information, where possible, about the remaining round trip energy efficiency as information on the state of health of the battery The update frequency of remaining round trip energy efficiency should be aligned with the update frequency of round trip energy efficiency fade and should be provided upon change of the battery status. DIN DKE Spec 99100 chapter reference: 6.7.4.4	⌈ 2 elements	0..1
[SMC] StateOfCharge	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#stateOfCharge Available capacity in a battery expressed as a percentage of remaining capacity.	⌈ 2 elements	1

3.2. Properties of the SMC "EnergyThroughput"

Figure 3 shows the UML-diagram for [SMC EnergyThroughput](#).

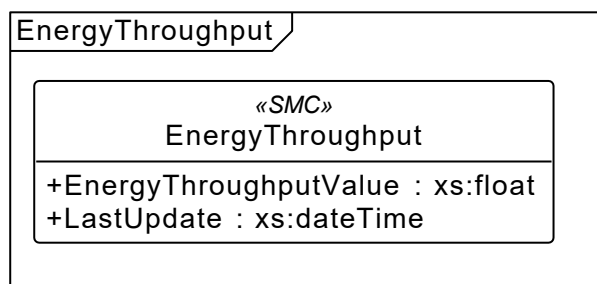


Figure 3. UML-Diagram for SMC "energyThroughput"

Data points related to energy throughput.

idShort:	EnergyThroughput		
Class:	SubmodelElementCollection		
semanticId:	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#energyThroughput		
Parent:	ProductCondition		
Explanation:			
Element details:	-		
[SME type]	semanticId	[valueType]	card.
idShort	Description@en	example	

[Prop] EnergyThroughputValue	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#energyThroughputValue supplementalSemanticId: urn:irdi:0173-1#02-ABL850#001 Overall sum of the energy throughput over the battery lifetime (data attribute should be reported as measured for further potential processing. In addition, the normalisation by usable battery energy could add a further useful value that ensures comparability among battery sizes).	[Float] 100000	1
[Prop] LastUpdate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#lastUpdate supplementalSemanticId: urn:irdi:0173-1#02-ABG740#003 Timestamp for dynamic data attributes show the last update time. Defined time within a time range, measured from the start time of the time range.	[DateTime] 2000-01-01T14:23:00	1

3.3. Properties of the SMC "CapacityThroughput"

Figure 4 shows the UML-diagram for [SMC capacityThroughput](#).

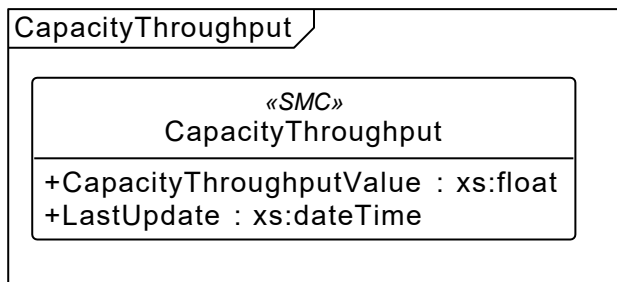


Figure 4. UML-Diagram for SMC "capacityThroughput"

Data points related to capacity throughput.

idShort:	CapacityThroughput		
Class:	SubmodelElementCollection		
semanticId:	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#capacityThroughput		
Parent:	ProductCondition		
Explanation:			
Element details:	-		
[SME type]	semanticId	[valueType]	card.
idShort	Description@en	example	

[Prop] CapacityThroughputValue	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#capacityThroughputValue supplementalSemanticId: urn:irdi:0173-1#02-ABL849#001 Overall sum of the capacity throughput over the battery lifetime (data attribute should be reported as measured for further potential processing. In addition, the normalisation by capacity could add a further useful value that ensures comparability among battery sizes).	[Float] 214	1
[Prop] LastUpdate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#lastUpdate supplementalSemanticId: urn:irdi:0173-1#02-ABG740#003 Timestamp for dynamic data attributes show the last update time. Defined time within a time range, measured from the start time of the time range.	[DateTime] 2000-01-01T14:23:00	1

3.4. Properties of the SMC "NumberOfFullCycles"

Figure 5 shows the UML-diagram for [SMC numberOfFullCycles](#).

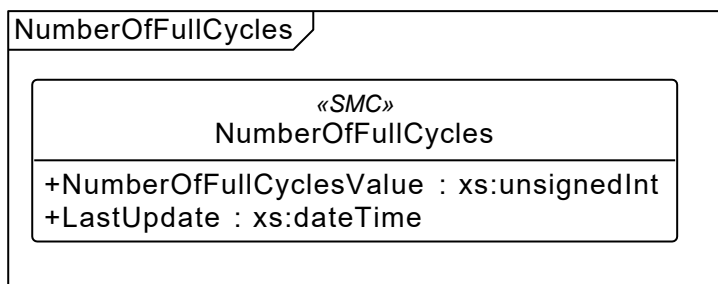


Figure 5. UML-Diagram for SMC "numberOfFullCycles"

Data points related to number of full charging and discharging cycles.

idShort:	NumberOfFullCycles		
Class:	SubmodelElementCollection		
semanticId:	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#numberOfFullCycles		
Parent:	ProductCondition		
Explanation:			
Element details:	-		
[SME type]	semanticId	[valueType]	card.
idShort	Description@en	example	

[Prop] NumberOfFullCyclesValue	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#numberOfFullCyclesValue supplementalSemanticId: urn:irdi:0173-1#02-ABL868#001 Total count of times a battery has been fully charged from empty to full capacity and then fully discharged, used as an indicator of the battery's usage and lifespan.	[UnsignedInt] 600	1
[Prop] LastUpdate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#lastUpdate supplementalSemanticId: urn:irdi:0173-1#02-ABG740#003 Timestamp for dynamic data attributes show the last update time. Defined time within a time range, measured from the start time of the time range.	[DateTime] 2000-01-01T14:23:00	1

3.5. Properties of the SMC "StateOfCertifiedEnergy"

Figure 6 shows the UML-diagram for SMC `stateOfCertifiedEnergy`.

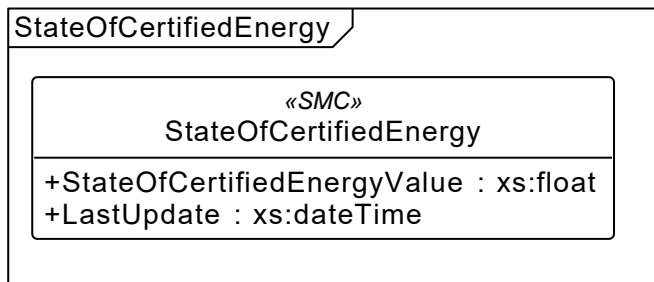


Figure 6. UML-Diagram for SMC "stateOfCertifiedEnergy"

Data points related to the state of certified energy.

idShort:	StateOfCertifiedEnergy		
Class:	SubmodelElementCollection		
semanticId:	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#stateOfCertifiedEnergy		
Parent:	ProductCondition		
Explanation:			
Element details:	-		
[SME type]	semanticId	[valueType]	card.
idShort	Description@en	example	

[Prop] StateOfCertifiedEnergyValue	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#stateOfCertifiedEnergyValue supplementalSemanticId: urn:irdi:0173-1#02-ABL847#001 Measured or on-board UBE performance at a specific point in its lifetime, expressed as a percentage of the certified usable battery energy.	[Float] 70	1
[Prop] LastUpdate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#lastUpdate supplementalSemanticId: urn:irdi:0173-1#02-ABG740#003 Timestamp for dynamic data attributes show the last update time. Defined time within a time range, measured from the start time of the time range.	[DateTime] 2000-01-01T14:23:00	1

3.6. Properties of the SMC "RemainingEnergy"

Figure 7 shows the UML-diagram for SMC *remainingEnergy*.

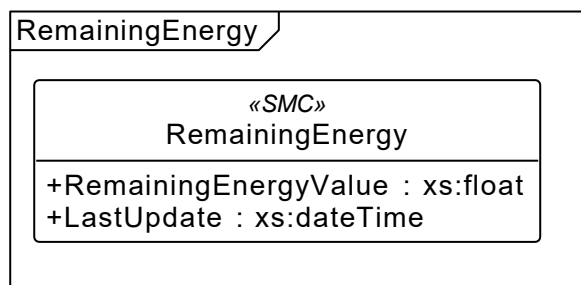


Figure 7. UML-Diagram for SMC "remainingEnergy"

Data points related to remaining energy.

idShort:	RemainingEnergy		
Class:	SubmodelElementCollection		
semanticId:	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#remainingEnergy		
Parent:	ProductCondition		
Explanation:			
Element details:	-		
[SME type]	semanticId	[valueType]	card.
idShort	Description@en	example	

[Prop] RemainingEnergyValue	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#remainingEnergyValue supplementalSemanticId: urn:irdi:0173-1#02-ABL870#001 UBE determined at the present point in the lifetime of the vehicle by the test procedure used for certification.	[Float] 30	1
[Prop] LastUpdate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#lastUpdate supplementalSemanticId: urn:irdi:0173-1#02-ABG740#003 Timestamp for dynamic data attributes show the last update time. Defined time within a time range, measured from the start time of the time range.	[DateTime] 2000-01-01T14:23:00	1

3.7. Properties of the SMC "RemainingCapacity"

Figure 8 shows the UML-diagram for SMC *remainingCapacity*.

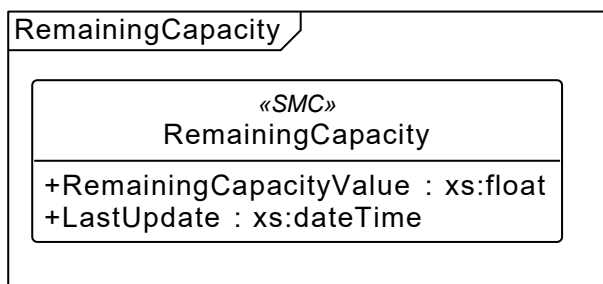


Figure 8. UML-Diagram for SMC "remainingCapacity"

Data points related to remaining capacity.

idShort:	RemainingCapacity		
Class:	SubmodelElementCollection		
semanticId:	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#remainingCapacity		
Parent:	ProductCondition		
Explanation:			
Element details:	-		
[SME type]	semanticId	[valueType]	card.
idShort	Description@en	example	

[Prop] RemainingCapacityValue	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#remainingCapacityValue supplementalSemanticId: urn:irdi:0173-1#02-ABL840#001 In-use data attribute on capacity, corresponding with the definition of rated capacity.	[Float] 56	1
[Prop] LastUpdate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#lastUpdate supplementalSemanticId: urn:irdi:0173-1#02-ABG740#003 Timestamp for dynamic data attributes show the last update time. Defined time within a time range, measured from the start time of the time range.	[DateTime] 2000-01-01T14:23:00	1

3.8. Properties of the SML "NegativeEvents"

Figure 9 shows the UML-diagram for [SML NegativeEvents](#).

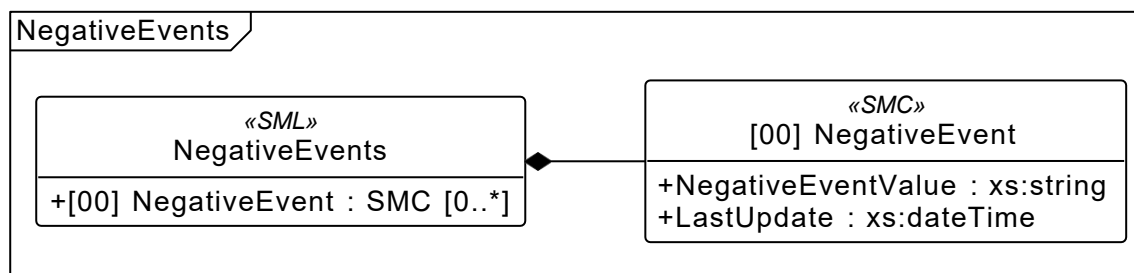


Figure 9. UML-Diagram for SML "negativeEvents"

Data points related to negative events.

idShort:	NegativeEvents		
Class:	SubmodelElementList		
semanticId:	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#negativeEvents		
Parent:	ProductCondition		
Explanation:			
Element details:	orderRelevant=No, typeValueListElement=SubmodelElementCollection		
[SME type]	semanticId	[valueType]	card.
idShort	Description@en	example	

[SMC] NegativeEvent	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#NegativeEvent The battery passport must contain information and data resulting from its use such as accidents. DIN DKE Spec 99100 chapter reference: 6.7.8.4	□ 2 elements	0..*
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3.9. Properties of the SML "InformationOnAccidents"

Figure 10 shows the UML-diagram for [SML InformationOnAccidents](#).

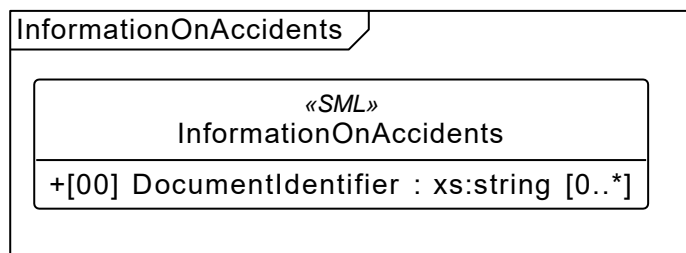


Figure 10. UML-Diagram for SML "InformationOnAccidents"

Data points related to a information on accidents.

idShort:	InformationOnAccidents		
Class:	SubmodelElementList		
semanticId:	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#informationOnAccidents		
Parent:	ProductCondition		
Explanation:			
Element details:	orderRelevant=No, typeValueListElement=Property		
[SME type]	semanticId	[valueType]	card.
idShort	Description@en	example	
[Prop] DocumentIdentifier	urn:samm:io.admin-shell.idta.handover_documentation:2.0.0#DocumentIdentifier Alphanumeric character sequence uniquely identifying a document.	[String] XF90-884	0..*

3.10. Properties of the SMC "TemperatureInformation"

Figure 11 shows the UML-diagram for [SMC TemperatureInformation](#).

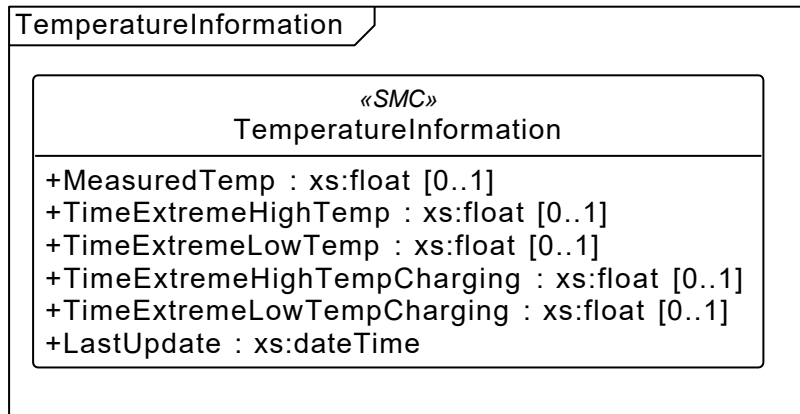


Figure 11. UML-Diagram for SMC "temperatureInformation"

Data points related to temperature.

idShort:	TemperatureInformation		
Class:	SubmodelElementCollection		
semanticId:	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#temperatureInformation		
Parent:	ProductCondition		
Explanation:			
Element details:	-		
[SME type]	semanticId	[valueType]	card.
idShort	Description@en	example	
[Prop] MeasuredTemp	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#measuredTemp Temperature measured in degree Celsius.	[Float] 20	0..1
[Prop] TimeExtremeHighTemp	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#timeExtremeHighTemp supplementalSemanticId: urn:irdi:0173-1#02-ABL816#001 Cumulated time spent above the given upper boundary of temperature.	[Float] 0	0..1
[Prop] TimeExtremeLowTemp	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#timeExtremeLowTemp supplementalSemanticId: urn:irdi:0173-1#02-ABL817#001 Cumulated time spent below the given lower boundary of temperature.	[Float] 0	0..1

[Prop] TimeExtremeHighTempCharging	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#timeExtremeHighTempCharging supplementalSemanticId: urn:irdi:0173-1#02-ABL818#001 Cumulated time spent above the given upper boundary of temperature during Charging.	[Float] 0	0..1
[Prop] TimeExtremeLowTempCharging	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#timeExtremeLowTempCharging supplementalSemanticId: urn:irdi:0173-1#02-ABL819#001 Cumulated time spent below the given lower boundary of temperature during charging.	[Float] 0	0..1
[Prop] LastUpdate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#lastUpdate supplementalSemanticId: urn:irdi:0173-1#02-ABG740#003 Timestamp for dynamic data attributes show the last update time. Defined time within a time range, measured from the start time of the time range.	[DateTime] 2000-01-01T14:23:00	1

3.11. Properties of the SMC "RemainingPowerCapability"

Figure 12 shows the UML-diagram for SMC *remainingPowerCapability*.

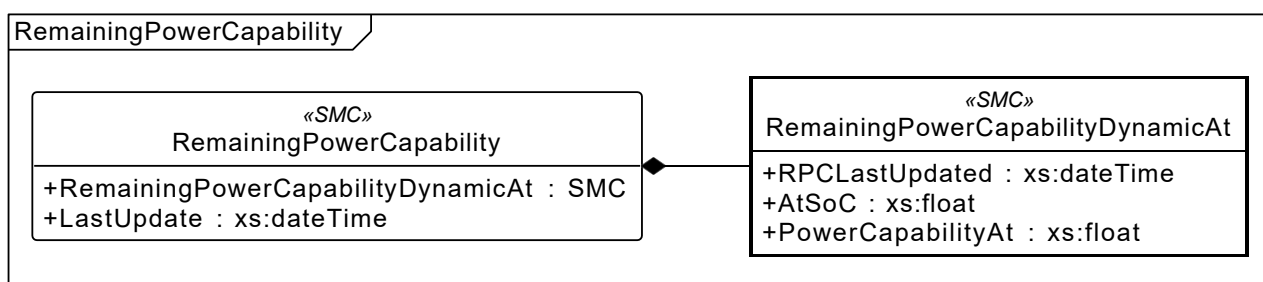


Figure 12. UML-Diagram for SMC "remainingPowerCapability"

Data points related to remaining power capability.

idShort:	RemainingPowerCapability
Class:	SubmodelElementCollection
semanticId:	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#remainingPowerCapability
Parent:	ProductCondition

Explanation:			
Element details:	-		
[SME type]	semanticId	[valueType]	card.
idShort	Description@en	example	
[SMC] RemainingPowerCapabilityDynamicAt	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#remainingPowerCapabilityDynamicAt supplementalSemanticId: urn:irdi:0173-1#02-ABL853#001 Defined by 'pre-use power capability (definition of power capability as given in Battery Regulation.	[] 3 elements	1
[Prop] LastUpdate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#lastUpdate supplementalSemanticId: urn:irdi:0173-1#02-ABG740#003 Timestamp for dynamic data attributes show the last update time. Defined time within a time range, measured from the start time of the time range.	[DateTime] 2000-01-01T14:23:00	1

3.12. Properties of the SMC "EvolutionOfSelfDischarge"

Figure 13 shows the UML-diagram for SMC *evolutionOfSelfDischarge*.

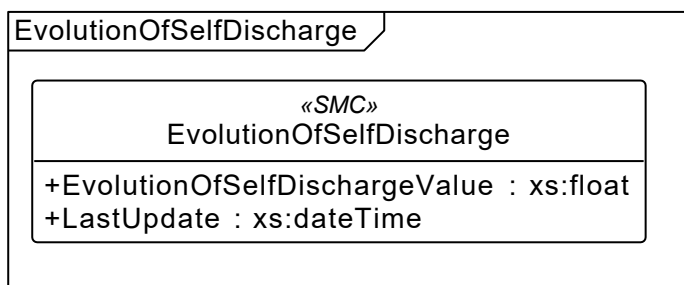


Figure 13. UML-Diagram for SMC "evolutionOfSelfDischarge"

Data points related to evolution of self discharge.

idShort:	EvolutionOfSelfDischarge
Class:	SubmodelElementCollection
semanticId:	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#evolutionOfSelfDischarge
Parent:	ProductCondition
Explanation:	

Element details:	-		
[SME type]	semanticId	[valueType]	card.
idShort	Description@en	example	
[Prop] EvolutionOfSelfDischargeValue	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#evolutionOfSelfDischargeValue supplementalSemanticId: urn:irdi:0173-1#02-ABL834#001 Total count of times a battery has been fully charged from empty to full capacity and then fully discharged, used as an indicator of the battery's usage and lifespan.	[Float] 1	1
[Prop] LastUpdate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#lastUpdate supplementalSemanticId: urn:irdi:0173-1#02-ABG740#003 Timestamp for dynamic data attributes show the last update time. Defined time within a time range, measured from the start time of the time range.	[DateTime] 2000-01-01T14:23:00	1

3.13. Properties of the SMC "CurrentSelfDischargingRate"

Figure 14 shows the UML-diagram for SMC `currentSelfDischargingRate`.

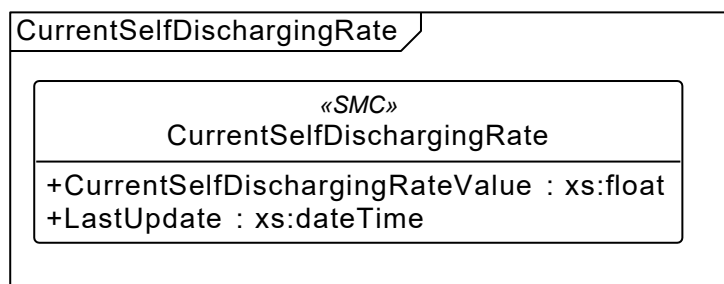


Figure 14. UML-Diagram for SMC "currentSelfDischargingRate"

Data points related to current self discharging rate.

idShort:	CurrentSelfDischargingRate
Class:	SubmodelElementCollection
semanticId:	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#currentSelfDischargingRate
Parent:	ProductCondition
Explanation:	

Element details:	-		
[SME type]	semanticId	[valueType]	card.
idShort	Description@en	example	
[Prop] CurrentSelfDischargingRateValue	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#currentSelfDischargingRateValue supplementalSemanticId: urn:irdi:0173-1#02-ABL825#001 Current self-discharge in % of capacity per unit of time in defined conditions (temperature range) during the use phase.	[Float] 2	1
[Prop] LastUpdate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#lastUpdate supplementalSemanticId: urn:irdi:0173-1#02-ABG740#003 Timestamp for dynamic data attributes show the last update time. Defined time within a time range, measured from the start time of the time range.	[DateTime] 2000-01-01T14:23:00	1

3.14. Properties of the SMC

"RemainingRoundTripEnergyEfficiency"

Figure 15 shows the UML-diagram for SMC *remainingRoundTripEnergyEfficiency*.

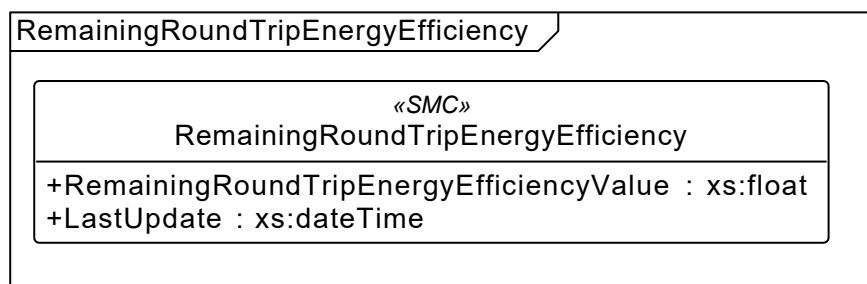


Figure 15. UML-Diagram for SMC "currentSelfDischargingRate"

Data points related to remaining round trip energy efficiency.

idShort:	RemainingRoundTripEnergyEfficiency
Class:	SubmodelElementCollection
semanticId:	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#remainingRoundTripEnergyEfficiency
Parent:	ProductCondition
Explanation:	

Element details:	-		
[SME type]	semanticId	[valueType]	card.
idShort	Description@en	example	
[Prop] RemainingRoundTripEnergyEfficiencyValue	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#remainingRoundTripEnergyEfficiencyValue supplementalSemanticId: urn:iridi:0173-1#02-ABL851#001 Percentage of energy that can be recovered and used after a full charge and discharge cycle, reflecting the battery's ability to retain and return energy efficiently over time.	[Float] 80	1
[Prop] LastUpdate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#lastUpdate supplementalSemanticId: urn:iridi:0173-1#02-ABG740#003 Timestamp for dynamic data attributes show the last update time. Defined time within a time range, measured from the start time of the time range.	[DateTime] 2000-01-01T14:23:00	1

3.15. Properties of the SMC "StateOfCharge"

Figure 16 shows the UML-diagram for SMC `stateOfCharge`.

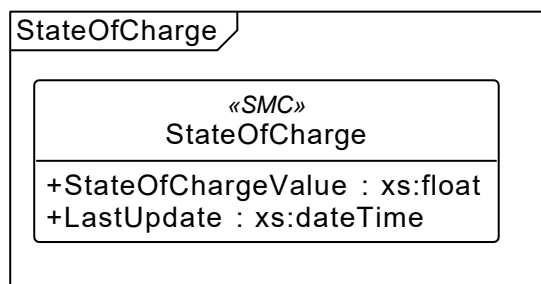


Figure 16. UML-Diagram for SMC "stateOfCharge"

Data points related to state of charge.

idShort:	StateOfCharge
Class:	SubmodelElementCollection
semanticId:	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#stateOfCharge
Parent:	ProductCondition
Explanation:	
Element details:	-

[SME type]	semanticId	[valueType]	card.
idShort	Description@en	example	
[Prop] StateOfCharge Value	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#stateOfChargeValue supplementalSemanticId: urn:irdi:0173-1#02-ABL821#001 Available capacity in a battery expressed as a percentage of remaining capacity.	[Float] 70	1
[Prop] LastUpdate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#lastUpdate supplementalSemanticId: urn:irdi:0173-1#02-ABG740#003 Timestamp for dynamic data attributes show the last update time. Defined time within a time range, measured from the start time of the time range.	[DateTime] 2000-01-01T14:23:00	1

3.16. Properties of the SMC "NegativeEvent"

Figure 17 shows the UML-diagram for SMC *NegativeEvent*.

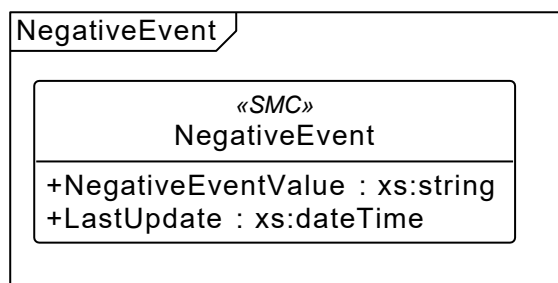


Figure 17. UML-Diagram for SMC "NegativeEvent"

Data points related to a negative event.

idShort:	NegativeEvent
Class:	SubmodelElementCollection
semanticId:	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#NegativeEvent
Parent:	NegativeEvents
Explanation:	The battery passport must contain information and data resulting from its use such as accidents. DIN DKE Spec 99100 chapter reference: 6.7.8.4
Element details:	-

[SME type]	semanticId	[valueType]	card.
idShort	Description@en	example	
[Prop] NegativeEvent Value	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#negativeEventValue Negative event, such as an accident. No further definition provided by regulation.	[String] overcharged	1
[Prop] LastUpdate	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#lastUpdate supplementalSemanticId: urn:irdi:0173-1#02-ABG740#003 Timestamp for dynamic data attributes show the last update time. Defined time within a time range, measured from the start time of the time range.	[DateTime] 2000-01-01T14:23:00	1

3.17. Properties of the SMC

"RemainingPowerCapabilityDynamicAt"

Figure 18 shows the UML-diagram for [SMC RemainingPowerCapabilityDynamicAt](#).

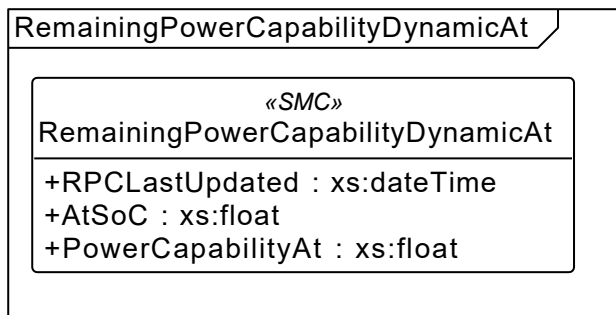


Figure 18. UML-Diagram for SMC "RemainingPowerCapabilityDynamicAt"

Data points of remaining power capability.

idShort:	RemainingPowerCapabilityDynamicAt
Class:	SubmodelElementCollection
semanticId:	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#remainingPowerCapabilityDynamicAt
Parent:	RemainingPowerCapability
Explanation:	
Element details:	-

[SME type]	semanticId	[valueType]	card.
idShort	Description@en	example	
[Prop] RPCLastUpdated	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#rPCLastUpdated RPC last updated.	[DateTime] 2000-01-01T14:23:00	1
[Prop] AtSoC	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#atSoC supplementalSemanticId: urn:irdi:0173-1#02-ABL821#001 Available capacity in a battery expressed as a percentage of remaining capacity.	[Float] 80	1
[Prop] PowerCapabilityAt	urn:samm:io.admin-shell.idta.batterypass.product_condition:1.0.1#powerCapabilityAt Power capability at.	[Float] 95	1

Annex A. Explanations on used table formats

1. General

The used tables in this document try to outline information as concise as possible. They do not convey all information on Submodels and SubmodelElements. For this purpose, the definitive definitions are given by a separate file in form of an AASX file of the Submodel template and its elements.

2. Tables on Submodels and SubmodelElements

For clarity and brevity, a set of rules is used for the tables for describing Submodels and SubmodelElements.

- The table heads abbreviate 'cardinality' with 'card'.
- The tables often place two informations in different rows of the same table cell. In this case, the first information is marked out by sharp brackets [] from the second information.
- The types of SubmodelElements are abbreviated (see [Table 1](#)):

Table 1. Abbreviations for SubmodelElements

SME type	SubmodelElement type
Blob	Blob
Cap	Capability
Ent	Entity
Evt	Event
File	File
MLP	MultiLanguageProperty
Opr	Operation
Prop	Property
Range	Range
Ref	ReferenceElement
Rel	RelationshipElement
RelA	AnnotatedRelationshipElement
SMC	SubmodelElementCollection
SME	SubmodelElement
SML	SubmodelElementList

- Multi-language strings are represented by the text value, followed by '@'-character and the ISO 639 language code: example@EN.
- The [valueType] is only given for Properties.

Annex B. Changes to the submodel template

General

This annex lists the changes from version to version of the Submodel, together with major changes in the overall document.

Changes Version 1.0.1

Major changes:

- new optional property "measuredTemp" as part of TemperatureInformation (<https://github.com/admin-shell-io/smt-semantic-models/issues/80>)

Changes Version 1.0

- First Version conformant to DIN DKE SPEC 99100

Bibliography

- [1] DIN DKE SPEC 99100, "Requirements for data attributes of the battery passport". February 2025.
- [2] "OMG Unified Modeling Language (OMG UML)", Formal/2017-12-05, Version 2.5.1. December 2018. [Online] Available: <https://www.omg.org/spec/UML/>
- [3] "Specification of the Asset Administration Shell", Publisher: Industrial Digital Twin Association (IDTA). [Online]. Available: <https://industrialdigitaltwin.org/en/content-hub/aasspecifications>
- [4] "Submodel Templates", Publisher: Industrial Digital Twin Association (IDTA). [Online]. Available: <https://industrialdigitaltwin.org/en/content-hub/submodels>
- [5] "How-to create a Submodel Template Specification", Publisher: Industrial Digital Twin Association (IDTA). June 2025. V1.1. [Online]. Available: https://industrialdigitaltwin.org/en/wp-content/uploads/sites/2/2025/06/IDTA_How-to-write-a-SMT-v1.1.pdf
- [6] "Semantic Aspect Meta Model (SAMM)", V2.2.0. [Online]. Available: <https://eclipse-esmf.github.io/samm-specification/2.2.0/index.html>
- [7] "Semantic Aspect Models - smt-semantic-models", Publisher: Industrial Digital Twin Association (IDTA). [Online]. Available: <https://github.com/admin-shell-io/smt-semantic-models>
- [8] "Semantic Aspect Models - BatteryPassDataModel", Publisher: BatteryPass Consortium. [Online]. Available: <https://github.com/batterypass/BatteryPassDataModel>
- [9] "Semantic Aspect Models - Tractus-X - sldt-semantic-models", [Online]. Publisher: Catena-X. [Online]. Available: <https://github.com/eclipse-tractusx/sldt-semantic-models>